

Software Requirements Specification (SRS) Template

Project: Healthcare Appointment & Patient Record System

Version: 1.0

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Approvals

Role	Name	Signature / Email	Date
Course Coordinator			

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1. Introduction

1.1 Purpose

This document specifies the functional, non-functional, security, interface, and verification requirements of the Healthcare Appointment & Patient Record System.

1.2 Scope

The system manages patients, doctors, appointments, medical histories, and prescriptions. It provides secure role-based access and prevents appointment conflicts. It covers authentication, authorization, medical-record management, and administration.

1.3 Audience

The system is intended for:

- Patients
- Doctors
- Receptionists
- Administrators
- Developers and QA engineers
- Project evaluators

1.4 Definitions

Term	Meaning
SRS	Software Requirements Specification
UI	User Interface
API	Application Programming Interface
RBAC	Role-Based Access Control
Passwor	Credential used for user authentication
EHR	Electronic Health Record
EMR	Electronic Medical Record

2. Overall description

2.1 Product perspective

The system is a centralized healthcare management application connecting patients, doctors, receptionists, and administrators. It manages appointments and patient records through a secure application and database.

2.2 Major product functions (detailed)

- User authentication and role-based access
- Patient and doctor management
- Doctor availability management
- Appointment booking, rescheduling, and cancellation
- Appointment conflict prevention
- Medical history management
- Prescription management
- Administrative monitoring and reporting

2.3 User roles and characteristics (expanded)

<u>Role</u>	<u>Main Responsibilities</u>
Patient	Manage profile, view doctors, book appointments, view records and prescriptions
Doctor	Manage availability, view appointments, update medical history and prescriptions
Receptionist	Register patients and manage appointments
Administrator	Manage users, roles, doctors, and system records

2.4 Operating environment

The system operates as a software application using a client interface, application server, database server, and secure network connection. Users can access the system through supported computing devices and web browsers.

2.5 Constraints

- Role-based access shall restrict users to authorized functions and records.
- Patient and medical information shall be protected from unauthorized access.
- Appointment conflicts shall be prevented.
- Data integrity shall be maintained for appointments and medical records.
- Authentication shall be required for protected system functions.

3. External interface requirements

3.1 User interfaces

Primary UI: Web-based interface with a clear and role-specific menu structure.

Accessibility: Simple navigation, readable text, and appropriate contrast for users with accessibility needs.

3.2 Hardware interfaces

No specialized hardware is required. The system shall operate on standard computers and devices with network connectivity.

3.3 Software interfaces

Database: Stores patient, doctor, appointment, medical history, and prescription data.

Backend API: Handles application logic, data processing, Authentication and management.

3.4 Communications

Secure Communication: HTTPS with **TLS 1.2 or higher** shall be used for communication between users and the system.

Data Exchange: The system shall securely exchange information between the application, backend, and database.

4. System features (detailed)

Each requirement below includes acceptance criteria and a reference test case. IDs follow HCS-F-####.

4.1 Authentication & Access Control

Req ID	Requirement	Type	Priority	Acceptance Criteria	Test
HCS-F-001	System shall authenticate users using registered email and password.	Functional	High	Valid credentials allow login; invalid credentials are rejected.	TC-001
HCS-F-002	System shall enforce role-based access for Patient, Doctor, Receptionist, and Administrator.	Functional	High	Users can access only permitted functions.	TC-002
HCS-F-003	System shall allow users to securely log out.	Functional	High	Session ends and protected pages require login again.	TC-003

4.2 Patient Management

Req ID	Requirement	Type	Priority	Acceptance Criteria	Test
HCS-F-004	Authorized users shall register and maintain patient information.	Functional	High	Patient data is saved and updated correctly.	TC-004
HCS-F-005	Patients shall view and update permitted profile information.	Functional	Medium	Allowed fields can be updated successfully.	TC-005

4.3 Doctor Management

Req ID	Requirement	Type	Priority	Acceptance Criteria	Test
HCS-F-006	Administrators shall create and maintain doctor profiles.	Functional	High	Doctor details can be added, edited, and viewed.	TC-006
HCS-F-007	Doctors shall manage their available appointment slots.	Functional	High	Slots can be added, modified, and removed.	TC-007

4.4 Appointment Management

Req ID	Requirement	Type	Priority	Acceptance Criteria	Test
HCS-F-008	Patients and receptionists shall book available appointments.	Functional	High	Appointment is created only for an available slot.	TC-008
HCS-F-009	System shall prevent conflicting appointments for the same doctor.	Functional	High	Duplicate time slots are rejected.	TC-009
HCS-F-010	Authorized users shall reschedule appointments to available slots.	Functional	High	Appointment changes to the selected valid slot.	TC-010
HCS-F-011	Authorized users shall cancel appointments.	Functional	Medium	Appointment status changes to Cancelled.	TC-011
HCS-F-012	System shall maintain appointment status.	Functional	Medium	Status shows Scheduled, Completed, Cancelled, or Rescheduled.	TC-012

4.5 Medical Records & Prescriptions

Req ID	Requirement	Type	Priority	Acceptance Criteria	Test
HCS-F-013	Authorized doctors shall add and update patient medical history.	Functional	High	Medical history is saved and updated correctly.	TC-013
HCS-F-014	Authorized users shall view medical records according to their role.	Functional	High	Only permitted records are displayed.	TC-014
HCS-F-015	Authorized doctors shall create and maintain prescriptions.	Functional	High	Prescription details are saved correctly.	TC-015
HCS-F-016	Patients shall view their medical records and prescriptions.	Functional	High	Patient can view only their own records.	TC-016

4.6 Administration & Monitoring

Req ID	Requirement	Type	Priority	Acceptance Criteria	Test
HCS-F-017	Administrators shall manage user accounts and roles.	Functional	High	Accounts and assigned roles can be managed.	TC-017
HCS-F-018	System shall record important system activities.	Functional	Medium	Changes include user and timestamp details.	TC-018

5.1. Security

5.1.1 Security Objectives

ID	Security Objective
SO-001	Protect patient and medical information from unauthorized access.

SO-002	Prevent unauthorized modification or deletion of appointments and medical records.
SO-003	Ensure users can access only the functions and data permitted by their assigned role.

5.1.2 Security Requirements

Req ID	Security Requirement	Priority	Acceptance Criteria	Test Case
HCS-S R-001	System shall require authentication before accessing protected functions and patient data.	High	Unauthenticated users cannot access protected pages.	TC-SR-001
HCS-S R-002	System shall enforce role-based access control for Patients, Doctors, Receptionists, and Administrators.	High	Users cannot access unauthorized functions.	TC-SR-002
HCS-S R-003	System shall use HTTPS with TLS 1.2 or higher for data transmission.	High	Communication uses HTTPS/TLS.	TC-SR-003
HCS-S R-004	System shall store passwords using secure one-way hashing and shall not store plaintext passwords.	High	Database contains only hashed passwords.	TC-SR-004
HCS-S R-005	System shall maintain audit logs for critical appointment, medical-record, and prescription changes.	Medium	Logs contain user and timestamp information.	TC-SR-005
HCS-S R-006	System shall prevent users from viewing or modifying patient records outside their authorization.	High	Unauthorized record access is denied.	TC-SR-006

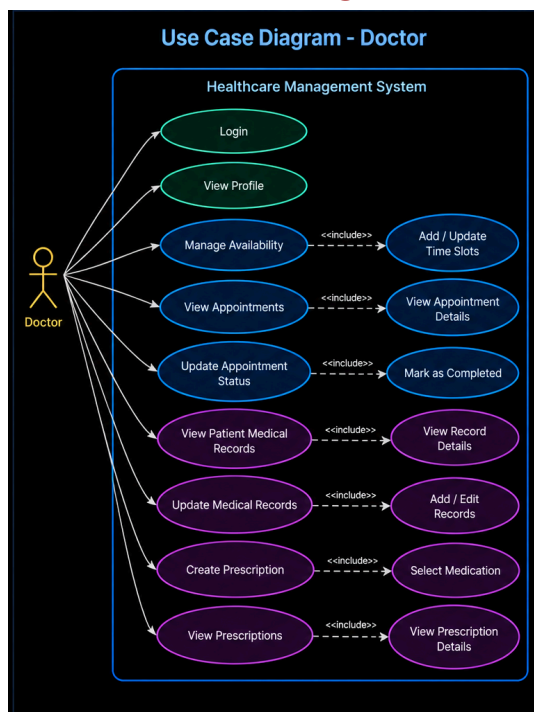
6. Quality attributes & Acceptance tests

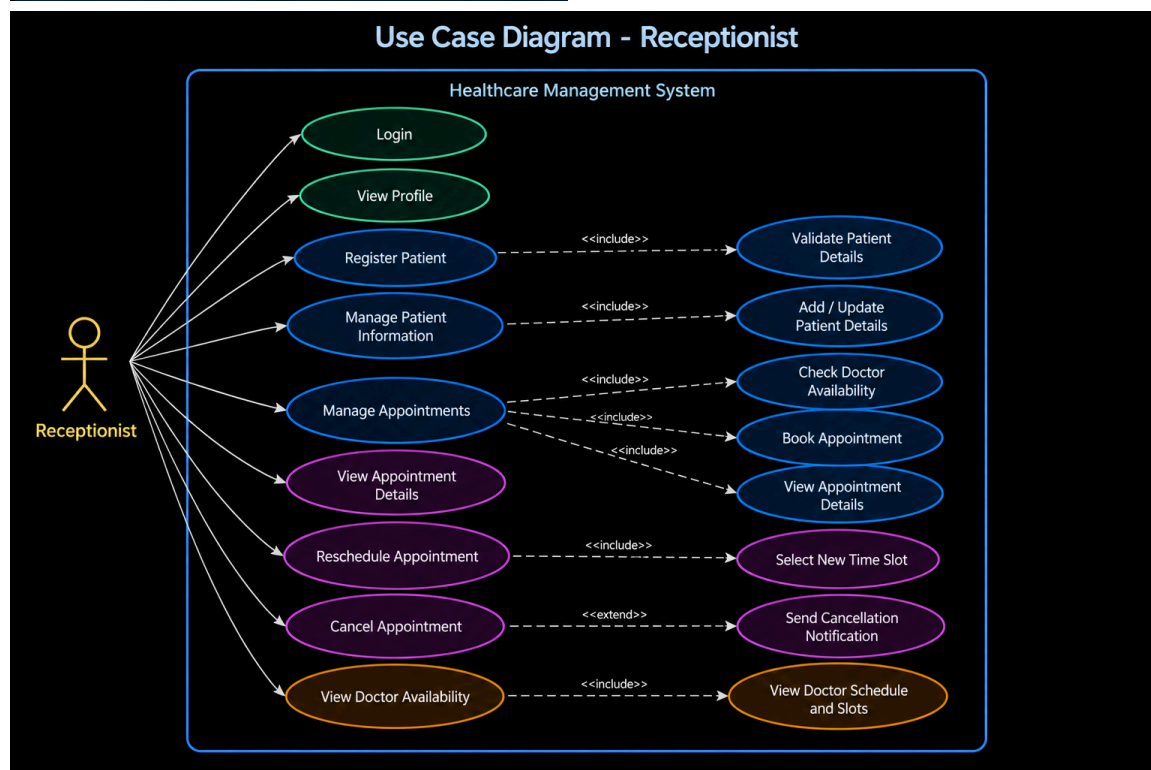
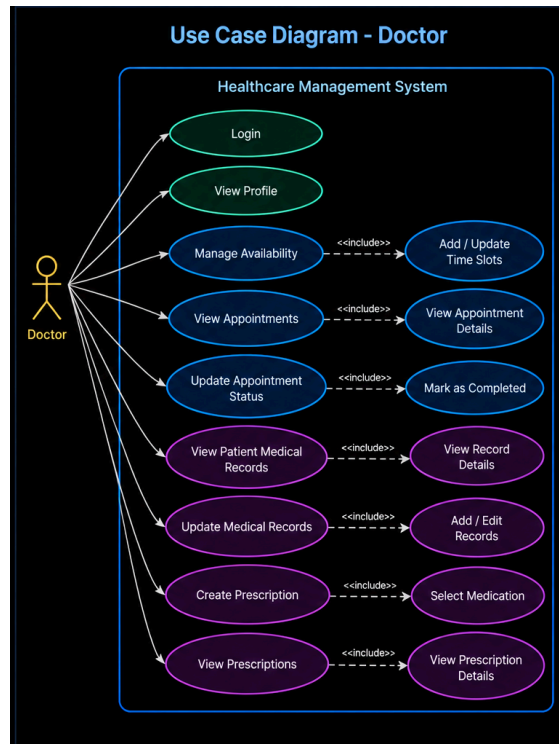
- **Quality attributes:** Performance, Availability, Security, Reliability, Usability, and Maintainability.

- **Acceptance test suites:** Authentication, Patient Management, Doctor Management, Appointment Management, Medical Records, Prescriptions, Performance, Security, and Usability.
- **Exit criteria for acceptance:** All high-priority functional requirements implemented and verified, no critical NFR or security failures, and RTM shows all required test cases passed.

7. System models and diagrams

7.1 UML Use-Case diagram





8. Requirements Traceability Matrix (RTM)

Req ID	Requirement	Section Ref / Design Spec	Module	Test Case(s)	Status	Comments
HCS-F-001	User authentication	4.1 / DS-Auth-01	Auth Module	TC-001	N	Email/password required
HCS-F-002	Role-based access	4.1 / DS-Auth-02	Access Module	TC-002	N	Patient, Doctor, Receptionist, Admin
HCS-F-003	Secure logout	4.1 / DS-Auth-03	Auth Module	TC-003	N	Session must end
HCS-F-004	Patient registration	4.2 / DS-Patient-01	Patient Module	TC-004	N	Authorized users
HCS-F-005	Patient profile update	4.2 / DS-Patient-02	Patient Module	TC-005	N	Permitted fields only
HCS-F-006	Doctor profile management	4.3 / DS-Doctor-01	Doctor Module	TC-006	N	Admin access
HCS-F-007	Doctor availability	4.3 / DS-Doctor-02	Doctor Module	TC-007	N	Manage time slots
HCS-F-008	Book appointment	4.4 / DS-Appt-01	Appointment Module	TC-008	N	Available slots only
HCS-F-009	Prevent appointment conflict	4.4 / DS-Appt-02	Appointment Module	TC-009	N	No duplicate slots
HCS-F-010	Reschedule appointment	4.4 / DS-Appt-03	Appointment Module	TC-010	N	New slot must be available
HCS-F-011	Cancel appointment	4.4 / DS-Appt-04	Appointment Module	TC-011	N	Status becomes Cancelled

HCS-F-012	Appointment status	4.4 / DS-Appt-05	Appointment Module	TC-012	N	Four status values
HCS-F-013	Update medical history	4.5 / DS-Med-01	Medical Record Module	TC-013	N	Doctor access
HCS-F-014	View medical records	4.5 / DS-Med-02	Medical Record Module	TC-014	N	Role-based access
HCS-F-015	Manage prescriptions	4.5 / DS-Pres-01	Prescription Module	TC-015	N	Doctor access
HCS-F-016	Patient views records	4.5 / DS-Med-03	Patient Module	TC-016	N	Own records only
HCS-F-017	User/role management	4.6 / DS-Admin-01	Admin Module	TC-017	N	Admin only
HCS-F-018	Activity logging	4.6 / DS-Admin-02	Audit Module	TC-018	N	User and timestamp recorded
HCS-NF-001	Response within 3 seconds	5 / DS-Perf-01	Core API / Web UI	TC-NF-001	N	Normal load
HCS-NF-002	System availability	5 / DS-Avail-01	System	TC-NF-002	N	Operating hours
HCS-NF-003	Data security	5 / DS-Sec-01	Security Module	TC-NF-003	N	Patient data protected
HCS-NF-004	Data reliability	5 / DS-Data-01	Database Module	TC-NF-004	N	No inconsistent records
HCS-NF-005	Ease of use	5 / DS-UI-01	Web UI	TC-NF-005	N	Simple navigation
HCS-NF-006	Maintainability	5 / DS-Maint-01	All Modules	TC-NF-006	N	Modular design

HCS-S R-001	Authentication required	5.2 / DS-Sec-01	Auth Module	TC-SR- 001	N	Protected functions
HCS-S R-002	Role-based access control	5.2 / DS-Sec-02	Access Module	TC-SR- 002	N	Unauthorized access denied
HCS-S R-003	TLS 1.2+	5.2 / DS-Sec-03	Communication	TC-SR- 003	N	HTTPS required
HCS-S R-004	Password hashing	5.2 / DS-Sec-04	Auth Module	TC-SR- 004	N	No plaintext passwords
HCS-S R-005	Audit logging	5.2 / DS-Sec-05	Audit Module	TC-SR- 005	N	Critical changes logged
HCS-S R-006	Patient record protection	5.2 / DS-Sec-06	Medical Record Module	TC-SR- 006	N	Unauthorized records blocked